

|                                                |                                                                 |
|------------------------------------------------|-----------------------------------------------------------------|
| Program : <b>Diploma in Textile Technology</b> |                                                                 |
| Course Code : <b>6067</b>                      | Course Title: <b>Fashion Design and Apparel Manufacture Lab</b> |
| Semester : <b>6</b>                            | Credits: <b>2.5</b>                                             |
| Course Category: <b>Program Elective</b>       |                                                                 |
| Periods per week: <b>4 (L:0, T:1, P:3)</b>     | Periods per semester: <b>60</b>                                 |

### Course Objectives:

- To introduce various class of garments provide thorough knowledge on garment measurements, pattern making and grading, spreading, pattern lay out and cutting operations.
- To impart thorough the knowledge on sewing operation, fusing machine, fusing operation, packing and packing material.
- To attain clear idea about product development and detailed process fashion designing using CAD.

### Course Prerequisites:

| Topic                  | Course code | Course name               | Semester |
|------------------------|-------------|---------------------------|----------|
| Computing fundamentals |             | Introduction to IT system | 1        |
| Fabric Manufacture     |             | FM- I & II                | 3,4      |

### Course Outcomes:

On completion of the course, the student will be able to:

| CO <sub>n</sub> | Description                                                                                             | Duration (Hours) | Cognitive Level |
|-----------------|---------------------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | Prepare pattern for various garments and apparels using the given measurement and do grading operation. | 15               | Applying        |
| CO2             | Practice the cutting operation for various garments and apparels with the pattern                       | 14               | Applying        |
| CO3             | Construct various garment and apparels using sewing and fusing machine.                                 | 15               | Applying        |
| CO4             | Illustrate the various operation related to apparel manufacturing with the aid of CAD.                  | 13               | Applying        |
|                 | Lab Exam                                                                                                | 3                |                 |

**CO-PO Mapping:**

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 3   |     |     |     |     |     |     |
| CO2             |     | 3   |     |     |     |     |     |
| CO3             |     |     | 3   |     |     |     |     |
| CO4             |     |     |     | 3   |     |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline:**

| Module Outcomes | Description                                                                                                                                                                                                          | Duration (Hours) | Cognitive Level |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|
| <b>CO1</b>      | <b>Prepare pattern for various garments and apparels using the given measurement and do grading operation</b>                                                                                                        |                  |                 |
| M1.01           | Prepare pattern Using the given measurement for Children Frock, Gathered Petticoat and Pyjama.                                                                                                                       | 4                | Applying        |
| M1.02           | Prepare pattern Using the given measurement for Slack Shirt, Bermudas and kurtha.                                                                                                                                    | 3                | Applying        |
| M1.03           | Prepare pattern Using the given measurement for Ladies skirt, Salwar & Kameez and kitchen apron                                                                                                                      | 4                | Applying        |
| M1.04           | Grade the shirt , kurtha and Ladies skirt to its next lower or higher sizes                                                                                                                                          | 4                | Applying        |
| <b>CO2</b>      | <b>Practice the cutting operation for various garments and apparels with the pattern</b>                                                                                                                             |                  |                 |
| M2.01           | Perform the cutting operation as per pattern for Children Frock, Gathered Petticoat and Pyjama                                                                                                                       | 5                | Applying        |
| M2.02           | Perform the cutting operation as per pattern for Slack Shirt, Bermudas and kurtha.                                                                                                                                   | 4                | Applying        |
| M2.03           | Perform the cutting operation as per pattern for Ladies skirt, Salwar & Kameez and kitchen apron.                                                                                                                    | 5                | Applying        |
|                 | Lab Exam -1                                                                                                                                                                                                          | 1.5              |                 |
| <b>CO3</b>      | <b>Construct various garments and apparels using sewing machine</b>                                                                                                                                                  |                  |                 |
| M3.01           | Familiarize with stitches and seams in different sewing machines and Construct different types of Seams, Hems, Plackets ,pockets, Button holes, Darts, Godets, pleats, Tucks, Frills, Shirring, Ruffles and Gathers. | 4                | Applying        |

|            |                                                                                                                                                               |     |          |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| M3.02      | Construct, finish and press Children Frock, Gathered Petticoat and Pyjama                                                                                     | 3   | Applying |
| M3.03      | Construct, finish and press Slack Shirt, Bermudas and kurtha.                                                                                                 | 4   | Applying |
| M3.04      | Construct, finish and press Ladies skirt, Salwar & Kameez and kitchen apron.                                                                                  | 4   | Applying |
| <b>CO4</b> | <b>Illustrate the various operation related to apparel manufacturing with the aid of CAD.</b>                                                                 |     |          |
| M4.01      | Familiarize with Garment CAD and use the screen for drafting by using the appropriate tools to set up drafting, editing and saving                            | 4   | Applying |
| M4.02      | Set up drafting patterns for shorts to a given size and grade it for the other sizes.                                                                         | 3   | Applying |
| M4.03      | Set up drafting a pattern for a trouser and grade it to the other size.                                                                                       | 3   | Applying |
| M4.04      | Estimate the material requirement by using layout. Compare the manual and computer grading in connection with time and labour. Set up printing through Bitmap | 3   | Applying |
|            | Lab Exam - II                                                                                                                                                 | 1.5 |          |

#### Text / Reference:

| T/R | Author/Book Title                                      |
|-----|--------------------------------------------------------|
| R1  | Taulkdar- Garment Technology- Wood head publishing Ltd |
| R2  | NIFT - Garment Manufacture - New Delhi                 |
| R3  | Carr.H - The technology of Clothing manufacture        |
| R4  | Berry N - Dress pattern designing                      |
| R5  | Mary Mathew- Practical Clothing Construction           |

#### Online Resources:

| Sl. No | Website Link                                                                                                                                      |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | <a href="http://www.nptel.ac.in/courses">http://www.nptel.ac.in/courses</a>                                                                       |
| 2      | <a href="http://freetextbooks.blogspot.in/2011/10/handbook-of-weaving.html">http://freetextbooks.blogspot.in/2011/10/handbook-of-weaving.html</a> |
| 3      | <a href="https://www.youtube.com/">https://www.youtube.com/</a>                                                                                   |
| 4      | <a href="http://textilelearner.blogspot.in">http://textilelearner.blogspot.in</a>                                                                 |
| 5      | <a href="https://www.texttronic.com">https://www.texttronic.com</a>                                                                               |